SD-WAN Configuration Documentation

HQ and Branch FortiGate Deployment

1. Network Topology Overview

The network consists of two sites:

Headquarters (HQ)

WAN Path

- Two routers provide dual WAN paths between sites:
    - **R1**: 10.10.10.1
    - **R2**: 10.20.20.1
- Both HQ and Branch FortiGate units connect to an IOU1 switch via VLAN trunk links.

R2

```
R2(config-if)#ip address 10.10.2.1 255.255.255.252
R2(config-if)#no sh
R2(config-if)#
*Nov 28 10:07:01.319: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state
o up
*Nov 28 10:07:02.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEther
et0/0, changed state to up
R2(config-if)#no ip address
R2(config-if)#exit
R2(config)#exit
R2#sh
*Nov 28 10:18:20.883: %SYS-5-CONFIG_I: Configured from console by console
R2#show sys int
         ^
% Invalid input detected at '^' marker.

R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 10.20.20.1 255.255.255.252
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
```

R1

```
et0/0, changed state to up
R1(config-if)#exit
R1(config)#int f0/1
                  ^
% Invalid input detected at '^' marker.

R1(config)#int f0/0
R1(config-if)#no ip address
R1(config-if)#exit
R1(config)#exit
R1#
*Nov 28 10:19:07.403: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.252
R1(config-if)#wr memory
              ^
% Invalid input detected at '^' marker.

R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
```

2. HQ FortiGate Configuration

2.1 WAN Interface Sub-Interfaces

Two VLAN-based WAN interfaces were created for the dual routers.

GUI Path:
`Network → Interfaces → Create New → Interface`

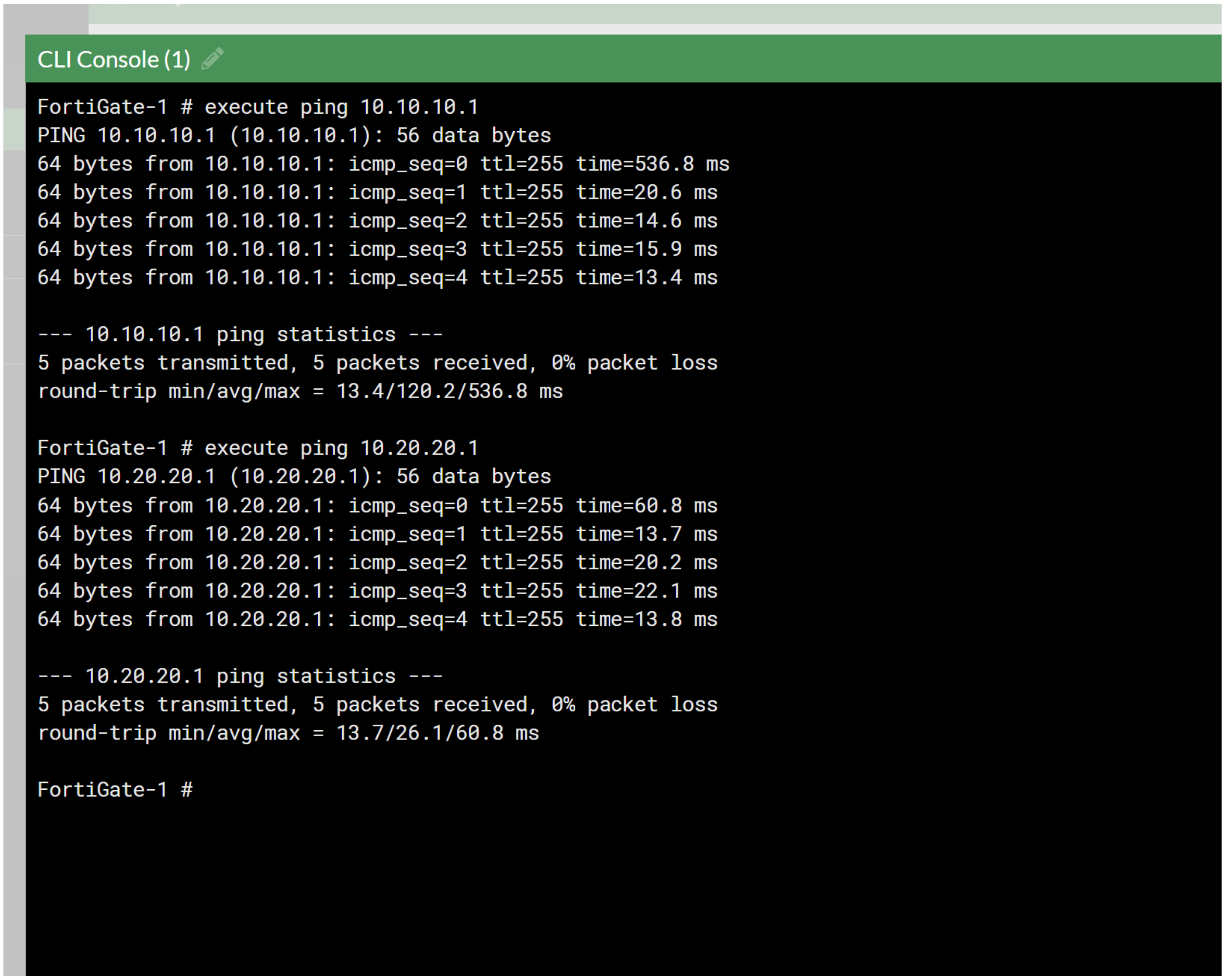
Parameter	WAN_R1	WAN_R2
Type	VLAN	VLAN
Parent Interface	port5	port5
VLAN ID	10	20
IP	10.10.10.2/30	10.20.20.2/30
Role	WAN	WAN
Access	PING	PING

CLI Equivalent

```
config system interface
    edit WAN_R1
        set vdom root
        set ip 10.10.10.2 255.255.255.252
        set allowaccess ping
        set role wan
        set interface port5
        set vlanid 10
    next
    edit WAN_R2
        set vdom root
        set ip 10.20.20.2 255.255.255.252
        set allowaccess ping
        set role wan
        set interface port5
        set vlanid 20
    next
end
```

Connectivity Test

```
execute ping 10.10.10.1  → Success
execute ping 10.20.20.1  → Success
```



2.2 SD-WAN Zone and Members

GUI Path:
`Network → SD-WAN → SD-WAN Zones`

Zone Name: `HQ_SD-WAN`

Members Added:

Member	Interfac	Gateway
1	WAN_R1	

		10.10.10.1
2	WAN_R2	10.20.20.1

⊟	SDWAN_ZONE		
•	port5	192.168.1.1	0
•	port2	192.168.1.2	0
•	WAN_R1	10.10.10.1	10
•	WAB_R2	10.20.20.1	20

2.3 Performance SLAs

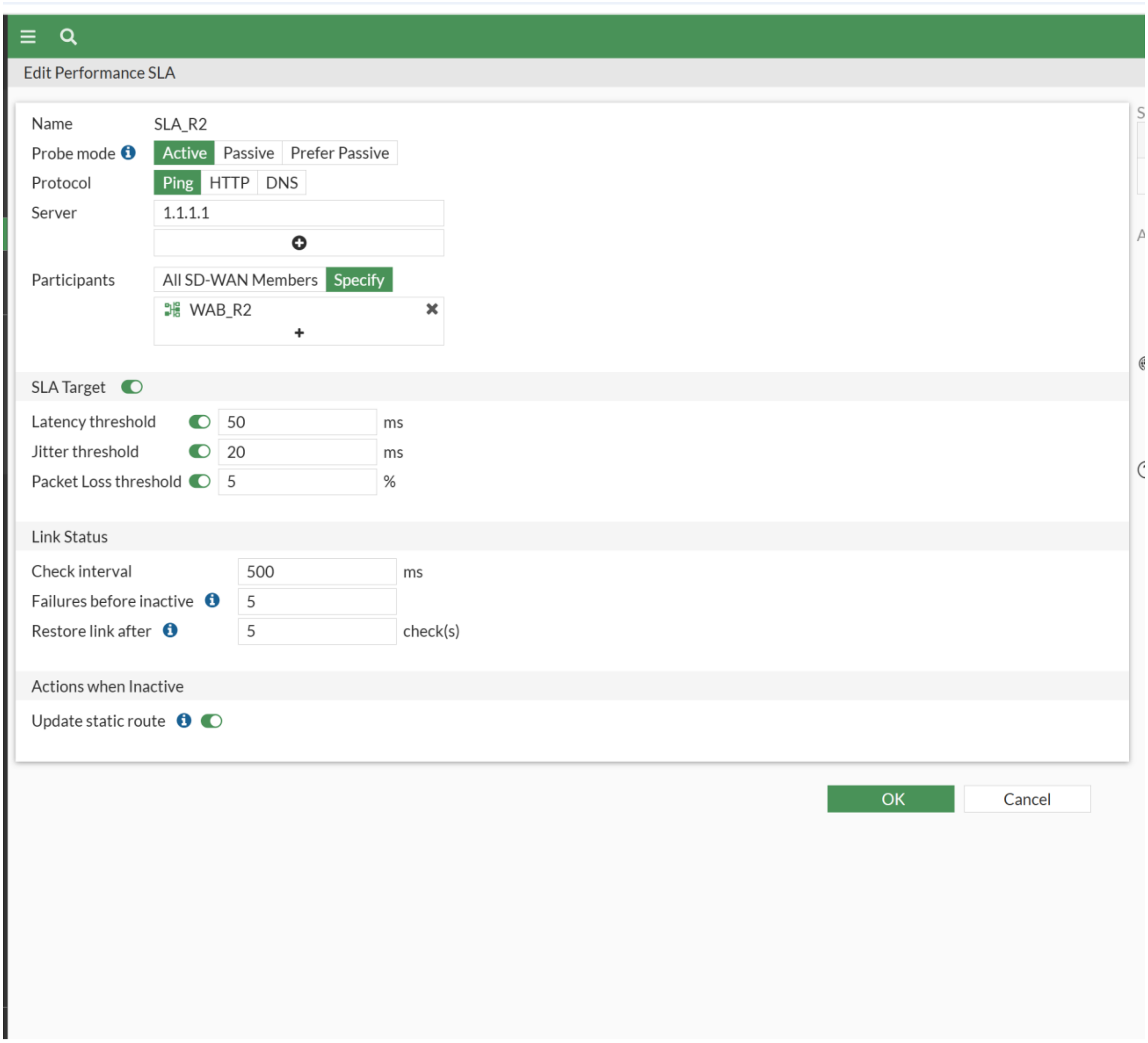
GUI Path:
`Network → SD-WAN → Performance SLAs`

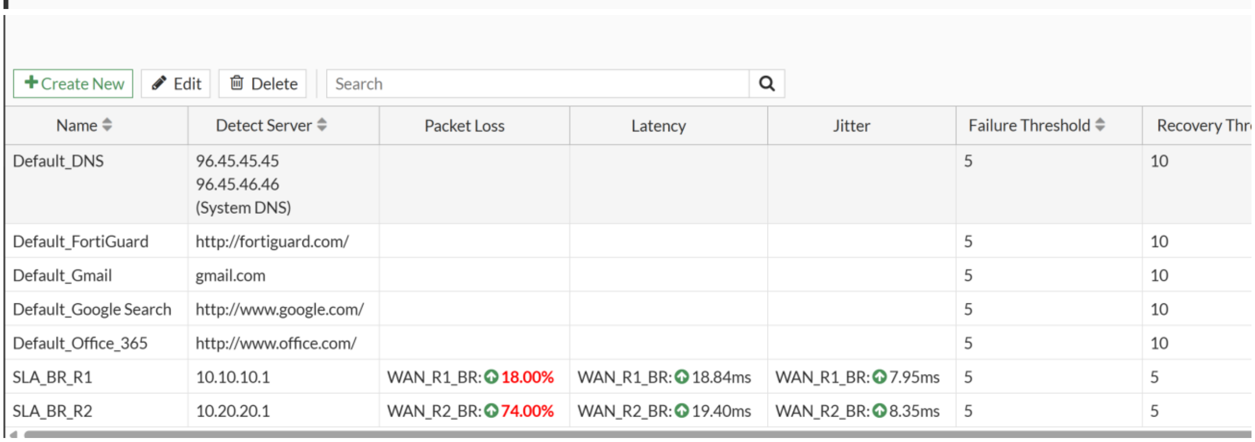
SLA Name	Monitored Interface	Probe Target	Metrics
SLA_R1	WAN_R1	8.8.8.8	Latency, Jitter, Packet Loss
SLA_R2	WAN_R2	1.1.1.1	Latency, Jitter, Packet Loss

Thresholds:

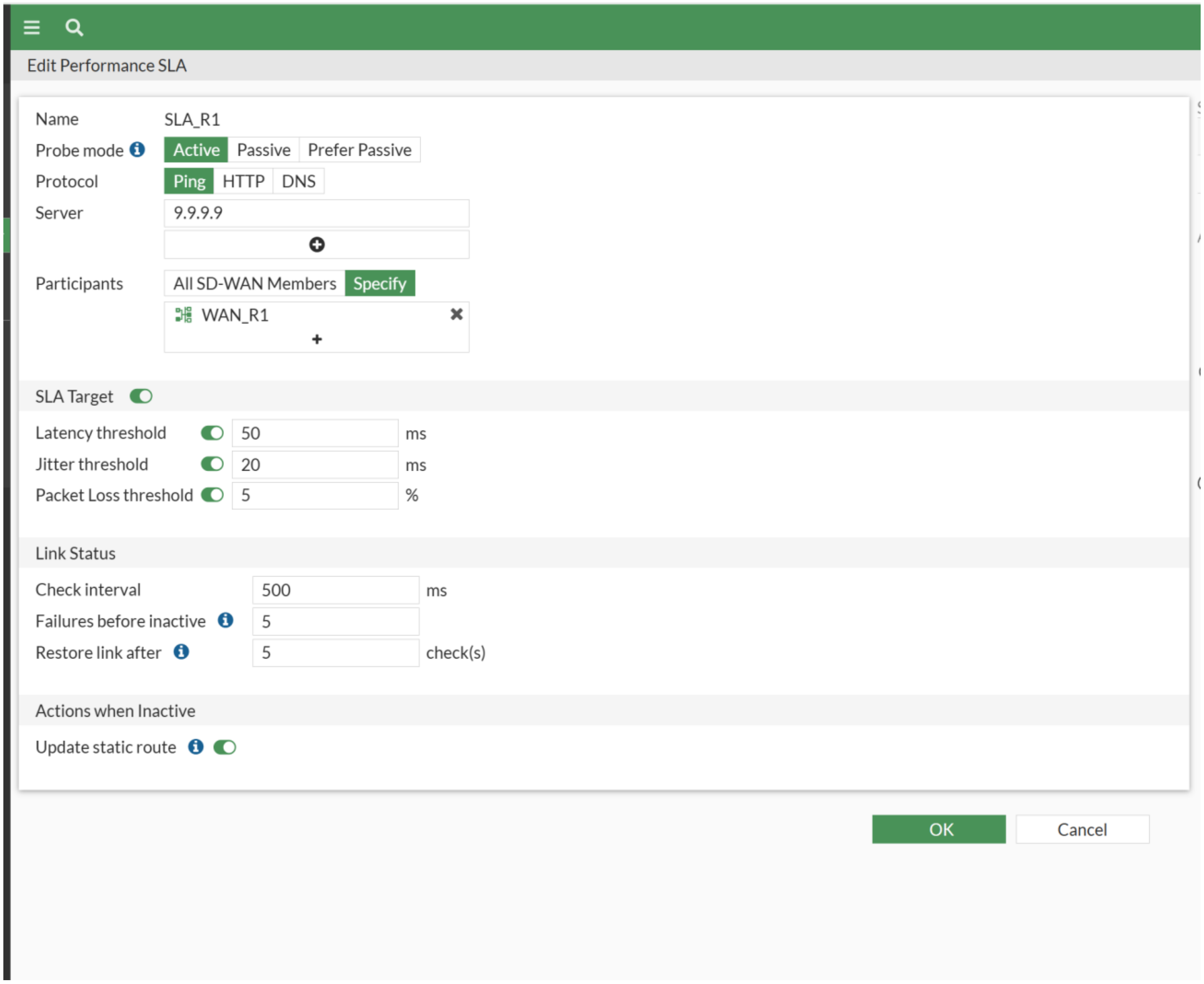
- Latency: **50 ms**
- Jitter: **20 ms**
- Packet Loss: **5%**

SLA status verified as UP.





Name	Detect Server	Packet Loss	Latency	Jitter	Failure Threshold	Recovery Thr
Default_DNS	96.45.45.45 96.45.46.46 (System DNS)				5	10
Default_FortiGuard	http://fortiguard.com/				5	10
Default_Gmail	gmail.com				5	10
Default_Google Search	http://www.google.com/				5	10
Default_Office_365	http://www.office.com/				5	10
SLA_BR_R1	10.10.10.1	WAN_R1_BR: 18.00%	WAN_R1_BR: 18.84ms	WAN_R1_BR: 7.95ms	5	5
SLA_BR_R2	10.20.20.1	WAN_R2_BR: 74.00%	WAN_R2_BR: 19.40ms	WAN_R2_BR: 8.35ms	5	5



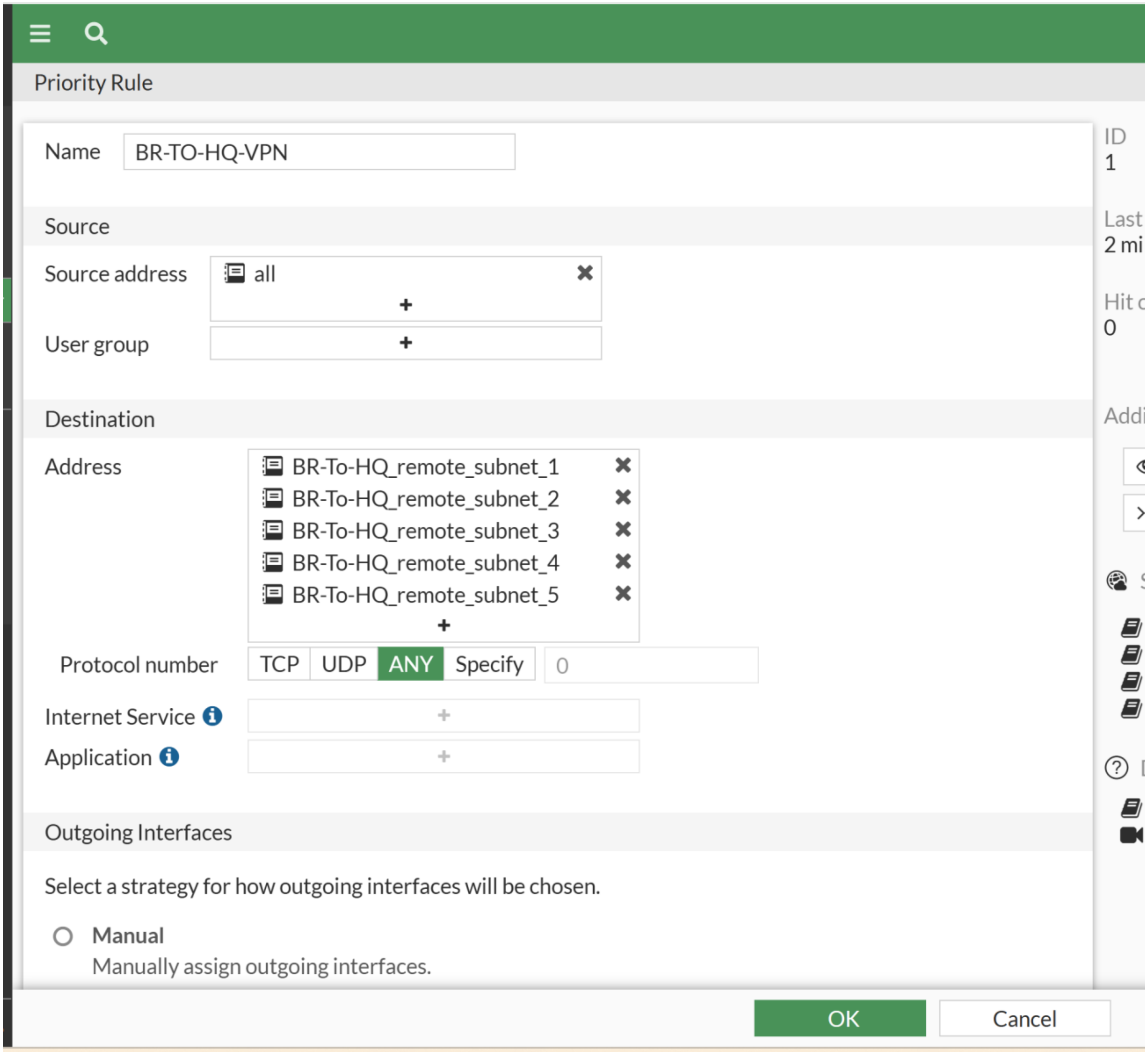
2.4 SD-WAN Rule

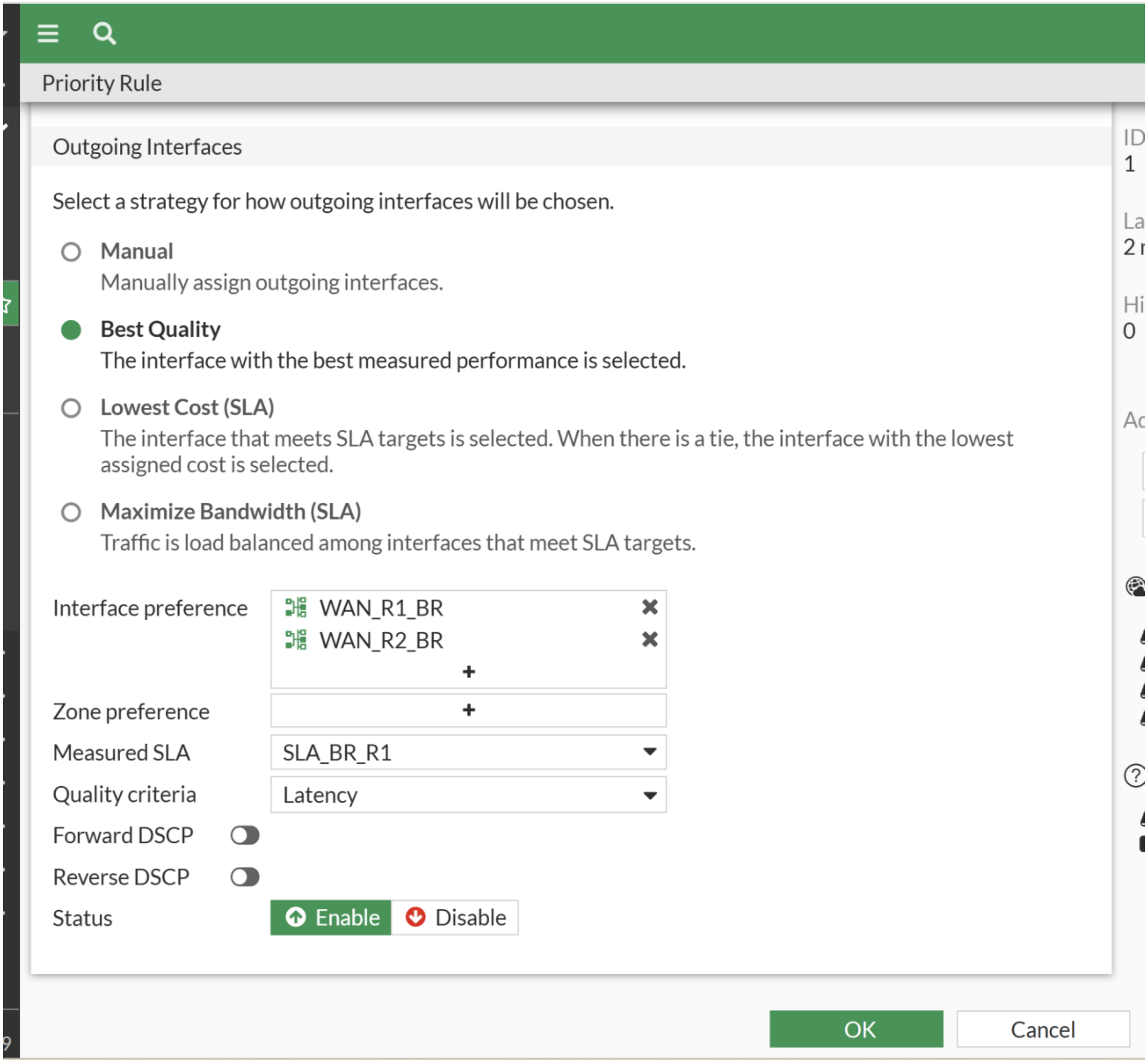
GUI Path:

`Network → SD-WAN → SD-WAN Rules`

Field	Value
Rule Name	HQ-BR-VPN
Source	all
Destination	Branch remote subnets (HQ-To-BR objects)
Strategy	Best Quality
Preferred Interfaces	WAN_R1, WAN_R2
Measured SLA	SLA_R1

Quality Criteria	Latency


Priority Rule
Name
BR-TO-HQ-VPN
ID
1
Last
2 mi
Hit c
0
Source
Source address
all
User group
Destination
Addi
Address
BR-To-HQ_remote_subnet_1
BR-To-HQ_remote_subnet_2
BR-To-HQ_remote_subnet_3
BR-To-HQ_remote_subnet_4
BR-To-HQ_remote_subnet_5
Protocol number
TCP
UDP
ANY
Specify
0
Internet Service
Application
Outgoing Interfaces
Select a strategy for how outgoing interfaces will be chosen.
Manual
Manually assign outgoing interfaces.
OK
Cancel



3. Branch FortiGate Configuration

Branch configuration mirrors HQ, using _BR suffixes.

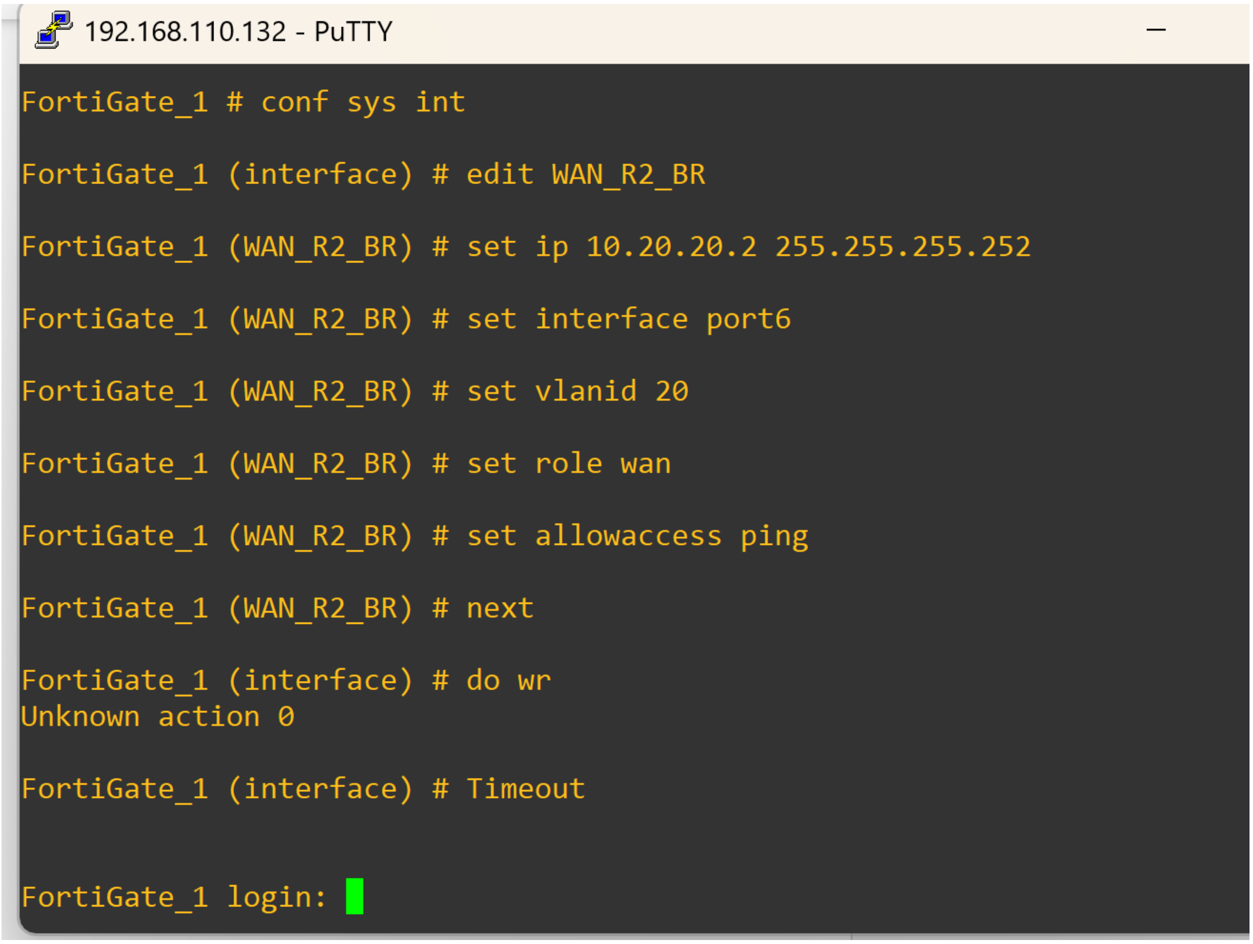
3.1 WAN Sub-Interfaces

Parameter	WAN_R1_BR	WAN_R2_BR
Type	VLAN	VLAN
Parent Interface	port6	port6
VLAN ID	10	20

IP	10.10.10.x/30	10.20.20.x/30
Role	WAN	WAN
Access	PING	PING

CLI

```
config system interface
    edit WAN_R1_BR
        set ip 10.10.10.x 255.255.255.252
        set vlanid 10
        set allowaccess ping
        set interface port6
    next
    edit WAN_R2_BR
        set ip 10.20.20.x 255.255.255.252
        set vlanid 20
        set allowaccess ping
        set interface port6
    next
end
```



3.2 SD-WAN Zone

Zone Name: BR_SD-WAN

Members:

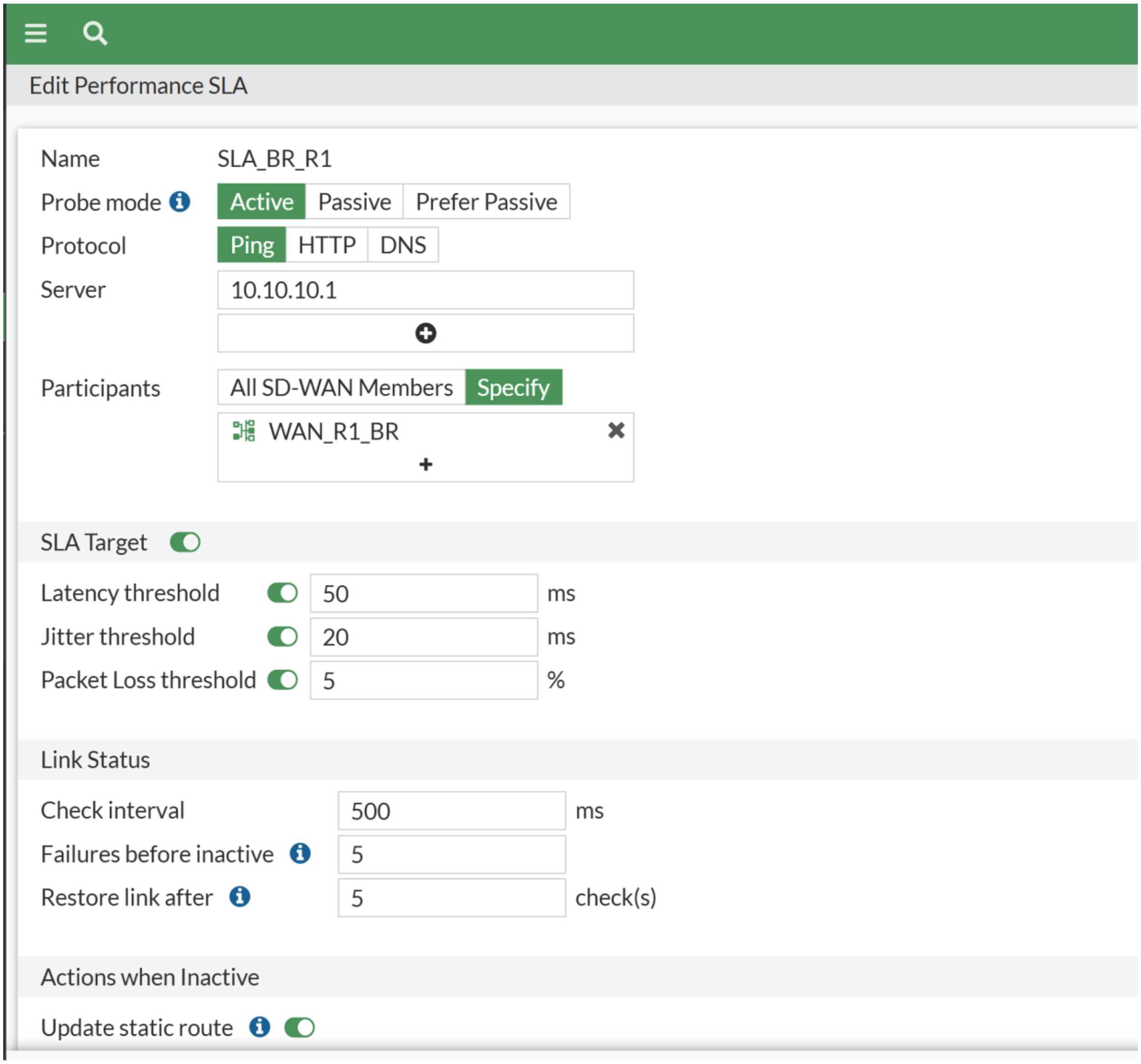
Interface	Gateway
WAN_R1_BR	10.10.10.1
WAN_R2_BR	10.20.20.1

3.3 Performance SLAs

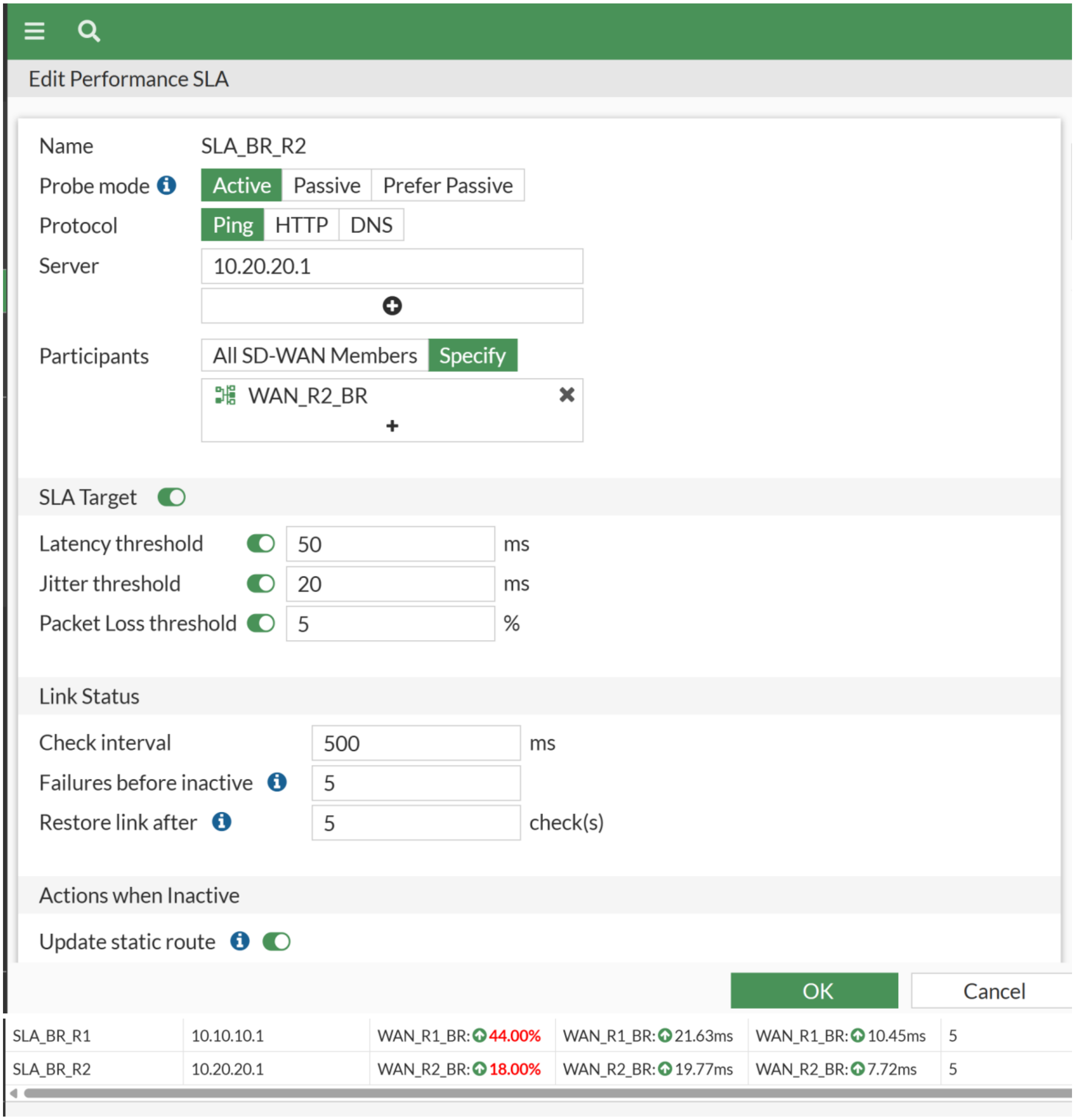
SLA Name	Interface	Target

SLA_BR_R1	WAN_R1_BR	10.10.10.1
SLA_BR_R2	WAN_R2_BR	10.20.20.1

Thresholds identical to HQ.



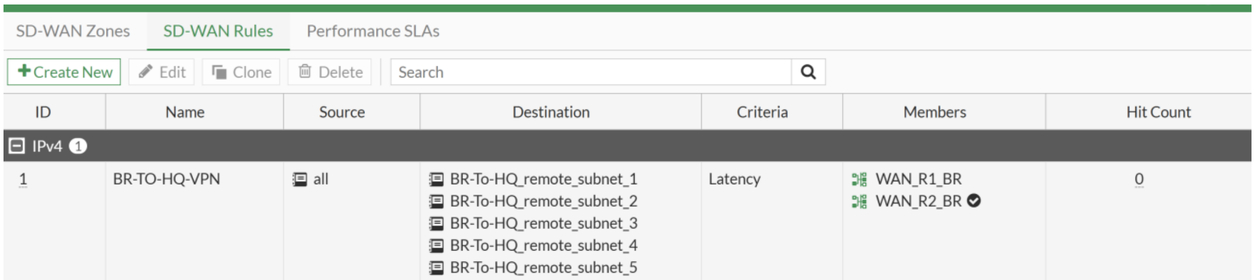
Edit Performance SLA
Name
SLA_BR_R1
Probe mode
Active
Passive
Prefer Passive
Protocol
Ping
HTTP
DNS
Server
10.10.10.1
Participants
All SD-WAN Members
Specify
WAN_R1_BR
SLA Target
Latency threshold
50
ms
Jitter threshold
20
ms
Packet Loss threshold
5
%
Link Status
Check interval
500
ms
Failures before inactive
5
Restore link after
5
check(s)
Actions when Inactive
Update static route



3.4 SD-WAN Rule

Field	Value
Name	BR-TO-HQ-VPN
Source	all
Destination	HQ remote subnets
Strategy	Best Quality

Preferred Interfaces	WAN_R1_BR, WAN_R2_BR
Measured SLA	SLA_BR_R1
Criteria	Latency



4. Status Summary

Component	HQ	Branch
WAN Interfaces	OK	OK
Gateway Reachability	OK	OK
SD-WAN Zone	Configured	Configured
SLA Monitoring	Active	Active
Rules	Applied	Applied